1 DESIGN BASELINE

SLAB SELECTION MATRIX

Slab	Top bars (X+Y)	Bottom bars (X+Y)	Ext. span	Int. span	Shear studs?
8 in	#5 @ 12 in O.C.	#4 @ 12 in O.C.	20’-0”	22’-0”	@ roof / rare
9 in	#5 @ 12 in O.C.*	#5 @ 12 in O.C.	22’-6”	24’-9”	yes
10 in	#5 @ 12 in O.C.*	#5 @ 12 in O.C.	25’-0”	27’-6”	yes
11 in	#5 @ 12 in O.C.*	#5 @ 12 in O.C.	27’-6”	30’-3”	yes
12 in	#5 @ 12 in O.C.*	#6 @ 12 in O.C.	30’-0”	33’-0”	yes

* BASE MAT; ADD COLUMN-STRIP TOP STEEL PER DESIGN.

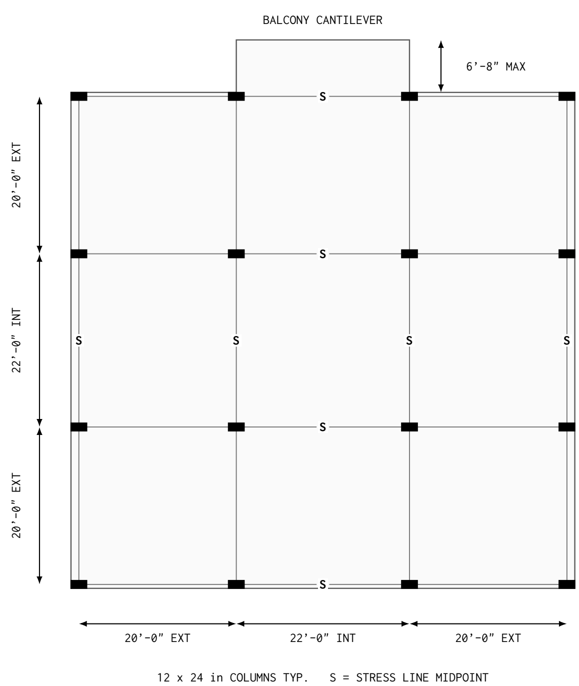
COLUMNS: 12 x 24 in DEFAULT; 10 in DIAMETER OR 10 x 30 in PROJECT-SPECIFIC.

LOAD BASIS

$$\begin{aligned} w_{\mathrm{sw},8} &= 100\text{ psf}, \quad w_{\mathrm{SDL}} = 35\text{ psf}, & D_8 &= w_{\mathrm{sw},8} + w_{\mathrm{SDL}} = 135\text{ psf} \\ L &= 40\text{ psf}, \quad w_{u,8} = 1.2D_8 + 1.6L = 226\text{ psf} \end{aligned} \tag{1}$$

$w_{\mathrm{sw},8}$ 8 in slab self-weight, 100 psf
w_{SDL} superimposed dead load, 35 psf
D_8 total unfactored dead load for the 8 in slab, 135 psf
$w_{u,8}$ factored typical-floor area load for the 8 in slab
L unfactored live area load, 40 psf

TYPICAL 3 BY 3 BAY LAYOUT



$$h \geq \frac{L_b}{10} \qquad \Rightarrow \qquad L_b \leq 10h = 80\text{ in} = 6\text{'-8"} \tag{2}$$

h slab thickness, 8 in
L_b balcony cantilever length

CONTROLLING BALCONY LIMIT: 6’-8” MAX

2 PROOF FOR GABE (TEMPORARY)

GOAL

Determine workable exterior and interior spans for 9, 10, 11, and 12 in conventionally reinforced plate slabs under confirmed SET criteria.

ACI 318-19 SPAN CONTROL

$$\ell_{n,e} \le 30h, \qquad \ell_{n,i} \le 33h, \qquad L_b \le 10h \tag{3}$$

$\ell_{n,e}$	clear exterior span without an edge beam [in]
$\ell_{n,i}$	clear interior span [in]
h	slab thickness [in]
L_b	balcony cantilever length [in]

LIMIT BASIS

Reference	ACI 318-19 Table 8.3.1.1; Grade 60; no drop panels
Exterior	no edge beam; $h \ge \ell_{n,e}/30$
Interior	$h \ge \ell_{n,i}/33$
Grid convention	listed grid span $\ge$ actual clear span; conservative

WORKABLE SPAN SCREEN

Slab	Exterior	Interior	Balcony	Status
8 in	20’-0”	22’-0”	6’-8”	accepted baseline
9 in	22’-6”	24’-9”	7’-6”	workable + studs
10 in	25’-0”	27’-6”	8’-4”	workable + studs
11 in	27’-6”	30’-3”	9’-2”	workable + studs
12 in	30’-0”	33’-0”	10’-0”	workable + studs

DEEPER-SLAB LOAD DEPENDENCY

$$w_{\mathrm{sw},h} = \gamma_c \left(\frac{h}{12\ \mathrm{in/ft}}\right), \qquad D_h = w_{\mathrm{sw},h} + w_{\mathrm{SDL}}, \qquad w_{u,h} = 1.2D_h + 1.6L \tag{4}$$

$w_{\mathrm{sw},h}$	slab self-weight for trial slab thickness h [psf]
D_h	unfactored dead area load for trial slab thickness h
w_{SDL}	superimposed dead load, 35 psf
γ_c	concrete unit weight, 150 pcf
$w_{u,h}$	factored area load for trial slab thickness h [psf]

FACTORED LOADS

Slab	$w_{\mathrm{sw},h}$	w_{SDL}	D_h	$w_{u,h}$
8 in	100.0 psf	35 psf	135.0 psf	226 psf
9 in	112.5 psf	35 psf	147.5 psf	241 psf
10 in	125.0 psf	35 psf	160.0 psf	256 psf
11 in	137.5 psf	35 psf	172.5 psf	271 psf
12 in	150.0 psf	35 psf	185.0 psf	286 psf

CONFIRMED BASIS

Code	ACI 318-19
Concrete strength	$f'_c = 5$ ksi
Steel strength	$f_y = 60$ ksi
Concrete unit weight	$\gamma_c = 150$ pcf; normalweight
Superimposed dead load	$w_{\mathrm{SDL}} = 35$ psf
Clear cover	$c_{\mathrm{clr}} = 0.75$ in

SHORING

Thickness trigger	none; slab thickness alone does not determine reshoring
12 in shoring case	150 psf fresh-concrete slab self-weight; add formwork and construction live loads
8 to 12 in slabs	shore during placement; reshore per engineered construction plan
Release criterion	verified in-place strength + construction-load capacity; not age alone
Required inputs	pour cycle; supported levels; slab age and strength; fresh concrete, formwork, worker, and equipment loads; shore spacing and stiffness
Reference	ACI PRC-347.2-17(25)

ACI DIRECT-DESIGN APPLICABILITY

Continuous spans	three in each direction; pass
Adjacent-span ratio	$L_i/L_e = 1.10 \leq 1.20$; pass
Panel aspect ratio	1.00 $\leq$ 2.00; pass
Column offset	zero; pass
Live-to-dead ratio	$L/D_8 = 0.30 \leq 2.00$; pass

INTERIOR-PANEL MOMENT SCREEN

$$\ell_n = L_i - 12\ \text{in}, \qquad \ell_2 = L_i, \qquad M_o = \frac{w_{u,h}\ell_2\ell_n^2}{8} \tag{5}$$

$$m_u^+ = \frac{2(0.60)(0.35)M_o}{\ell_2} = \frac{0.42M_o}{\ell_2}, \qquad m_u^- = \frac{2(0.75)(0.65)M_o}{\ell_2} = \frac{0.975M_o}{\ell_2} \tag{6}$$

L_e listed exterior grid span [ft]
L_i listed interior grid span [ft]
ℓ_n governing interior clear span using the 12 in column side [ft]
ℓ_2 transverse panel width [ft]
M_o total static factored moment [kip-ft]
m_u^+ positive column-strip factored moment per foot [kip-ft/ft]
m_u^- negative column-strip factored moment per foot [kip-ft/ft]

FLEXURAL STRENGTH

$$a = \frac{A_s f_y}{0.85 f_c' b}, \qquad \phi_f m_n = \frac{\phi_f A_s f_y}{12}\left(d_f - \frac{a}{2}\right), \qquad \phi_f = 0.90 \tag{7}$$

a equivalent compression-block depth [in]
A_s flexural reinforcement per foot [in^2/ft]
f_y reinforcement yield strength, 60 ksi
b design-strip width, 12 in
m_n nominal flexural strength per foot [kip-ft/ft]
d_f effective flexural depth to governing inner bar layer [in]
ϕ_f flexural strength-reduction factor, 0.90

POSITIVE-MOMENT BOTTOM MAT

Slab	m_u^+	Bottom mat	$\phi_f m_n$	DCR	$A_{s,\min}$
9 in	7.14	#5 @ 12 in O.C.	9.95	0.72	pass
10 in	9.44	#5 @ 12 in O.C.	11.34	0.83	pass
11 in	12.17	#5 @ 12 in O.C.	12.74	0.96	pass
12 in	15.38	#6 @ 12 in O.C.	19.54	0.79	pass

DCR $= m_u^+/(\phi_f m_n)$; $A_{s,\min} = 0.0018bh$; $A_{s,\min}$ = MINIMUM SLAB REINFORCEMENT PER FOOT.

NEGATIVE-MOMENT COLUMN STRIP

Slab	m_u^-	Trial top steel	A_s	$\phi_f m_n$	DCR
9 in	16.57	#5 @ 12 in O.C. + #4 @ 10 in O.C.	0.550	17.30	0.96
10 in	21.91	#5 @ 12 in O.C. + #5 @ 12 in O.C.	0.620	22.17	0.99
11 in	28.26	#5 @ 12 in O.C. + #6 @ 12 in O.C.	0.750	29.31	0.96
12 in	35.69	#5 @ 12 in O.C. + #7 @ 12 in O.C.	0.910	38.50	0.93

FLEXURE RESULT: PASS WITH LISTED MATS + TRIAL COLUMN-STRIP TOP STEEL

INTERIOR-COLUMN PUNCHING GEOMETRY

$$d_v = h - c_{\mathrm{clr}} - d_b, \qquad A_t = \left(\frac{L_e + L_i}{2}\right)^2, \qquad A_o = \frac{(C_x + d_v)(C_y + d_v)}{144} \tag{8}$$

$$V_u = w_{u,h}(A_t - A_o), \qquad b_o = 2(C_x + C_y + 2d_v), \qquad v_{u,0} = \frac{V_u}{b_o d_v} \tag{9}$$

d_v	average effective depth for punching [in]
c_{clr}	clear cover, 0.75 in
d_b	governing top support-bar diameter [in]
A_t	interior-column tributary area [ft^2]
A_o	area inside the critical perimeter [ft^2]
C_x, C_y	column dimensions, 12 and 24 in
V_u	direct factored shear across the critical perimeter [lb]
b_o	critical perimeter at $d_v/2$ from column faces [in]
$v_{u,0}$	direct factored punching stress [psi]

GRAVITY MOMENT TRANSFER

$$v_{u,\max} = v_{u,0} + \frac{\gamma_{vx} M_{ux} y_c}{J_{cx}} + \frac{\gamma_{vy} M_{uy} x_c}{J_{cy}} \tag{10}$$

$v_{u,\max}$	maximum factored punching stress [psi]
M_{ux}, M_{uy}	DDM gravity span-imbalance moments [lb-in]
γ_{vx}, γ_{vy}	fractions transferred by eccentricity of shear
x_c, y_c	critical-section centroid-to-perimeter distances [in]
J_{cx}, J_{cy}	critical-section properties for moment transfer [in^4]

CONCRETE PUNCHING STRENGTH

$$\lambda_s = \min\left(1, \sqrt{\frac{2}{1 + d_v/(10\ \mathrm{in})}}\right), \qquad \phi_v v_c = \phi_v\, 4\lambda_s \sqrt{f'_c}, \qquad \phi_v = 0.75 \tag{11}$$

λ_s	ACI size-effect factor, d_v in inches
v_c	nominal concrete punching strength [psi]
f'_c	specified concrete strength, 5000 psi
ϕ_v	shear strength-reduction factor, 0.75

PUNCHING SCREEN

Slab	d_v	$v_{u,0}$	$v_{u,\max}$	$\phi_v v_c$	DCR	Studs
8 in	6.625	151 psi	193 psi	212 psi	0.91	@ roof / rare
9 in	7.625	171 psi	222 psi	212 psi	1.05	yes
10 in	8.625	191 psi	252 psi	212 psi	1.19	yes
11 in	9.500	215 psi	288 psi	212 psi	1.36	yes
12 in	10.375	240 psi	327 psi	210 psi	1.56	yes

DCR $= v_{u,\max}/(\phi_v v_c)$; DEMAND/CAPACITY RATIO.

$$v_{u,\max} \le 8\phi_v \sqrt{f'_c} = 424\ \mathrm{psi} \tag{12}$$

STUD FEASIBILITY: PASS; MAXIMUM SCREENED DEMAND = 327 PSI

PROOF RESULT

Span/depth	pass per ACI 318-19 Table 8.3.1.1
Deflection	prescriptive thickness control used; calculation not required
Flexure	pass with listed bottom mats and trial top column-strip steel
Punching	9–12 in options require studs; inner compression cap passes
Result	listed 9–12 in exterior and interior spans are workable
Final design	size stud rails; check outer, edge, and corner critical sections

TEMPORARY PROOF — REMOVE AFTER GABE REVIEW